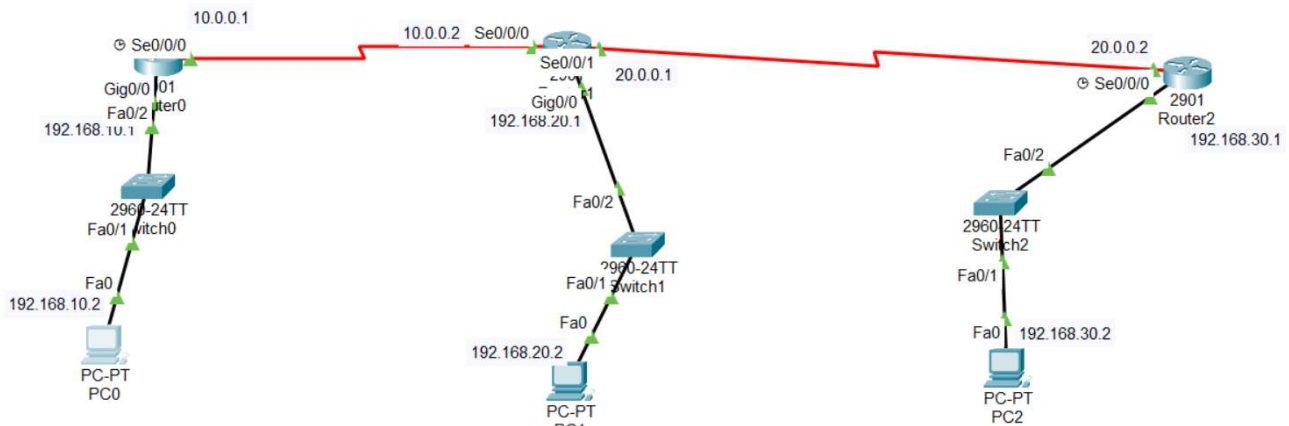


Day 3: Configuring RIP

Task 1: Set up a network topology with three routers in Cisco Packet Tracer, connect them using serial or Ethernet links, and assign IP addresses to all interfaces.



RIP (Routing Information Protocol) is a distance-vector dynamic routing protocol that decides the best path to a network purely by counting hops — the number of routers a packet must pass through to reach the destination. Each router adds 1 to the hop count for every network it advertises out an interface, so a directly connected network has a hop count (metric) of 0 from the router's own point of view, but shows as 1 hop to a neighbouring router that learns it.

RIP allows a maximum of 15 hops; a hop count of 16 is treated as “infinity”, meaning the network is unreachable.

Updates are how routers share this information: by default, every RIP router broadcasts (or multicasts, in RIPv2) its entire routing table to its directly connected neighbours every 30 seconds.

RIP also sends triggered updates immediately whenever a route changes, instead of waiting for the next scheduled 30-second cycle, which speeds up how quickly other routers find out about a change.

Task 2: Enable RIP version 2 on all three routers in your topology, advertise all directly connected networks, and verify that each router learns routes to all subnets using the 'show ip route' command.

RIP version 2 was enabled on each router with the following configuration pattern (shown here for Router0):

```
Router0(config)# router rip
Router0(config-router)# version 2
Router0(config-router)# network 192.168.10.0
Router0(config-router)# network 10.0.0.0
Router0(config-router)# no auto-summary
```

The same pattern (with each router's own directly connected networks) was applied on Router1 and Router2. show ip protocols was then used on each router to confirm RIP was active and check its version settings:

Router0

Physical | Config | CLI | Attributes |

IOS Command Line Interface

```
admin(config)#do wr
Building configuration...
[OK]
admin(config)#exit
admin#
%SYS-5-CONFIG_I: Configured from console by console

admin#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 24 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface          Send Recv Triggered RIP Key-chain
GigabitEthernet0/0   22
Serial0/0/0          22
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
  10.0.0.0
  20.0.0.0
  192.168.10.0
  192.168.20.0
Passive Interface(s):
Routing Information Sources:
  Gateway           Distance      Last Update
Distance: (default is 120)
admin#
```

Figure 1: Router0 – show ip protocols

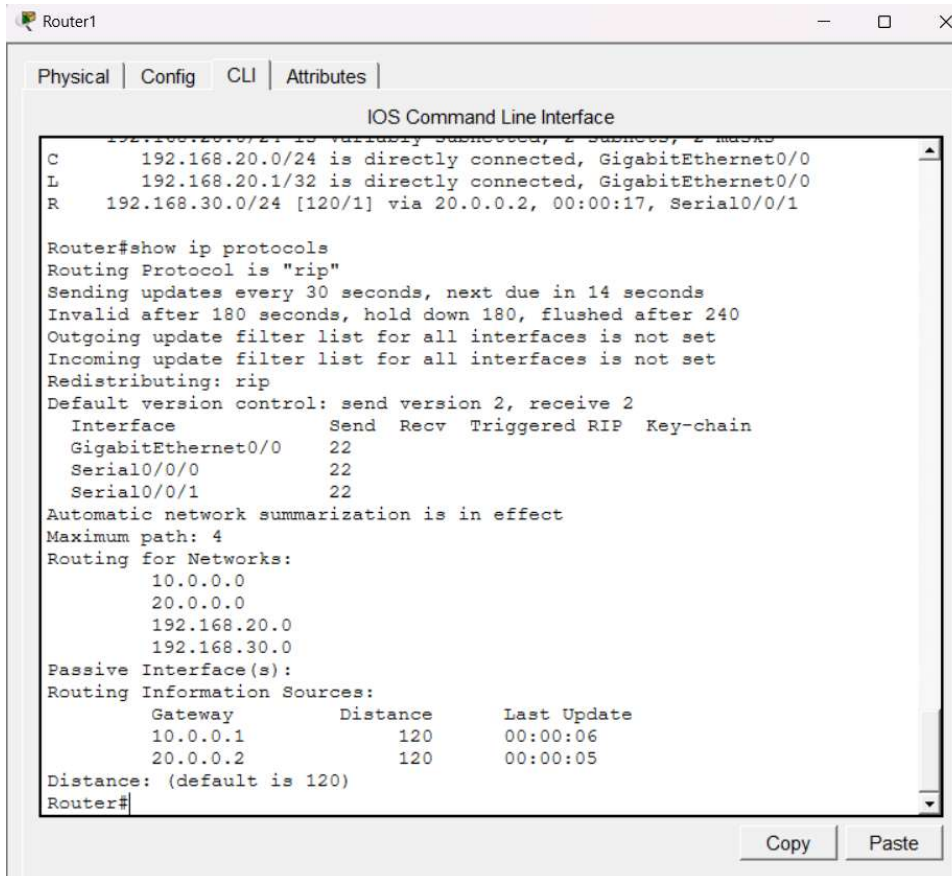


Figure 2: Router1 – show ip route (tail) and show ip protocols

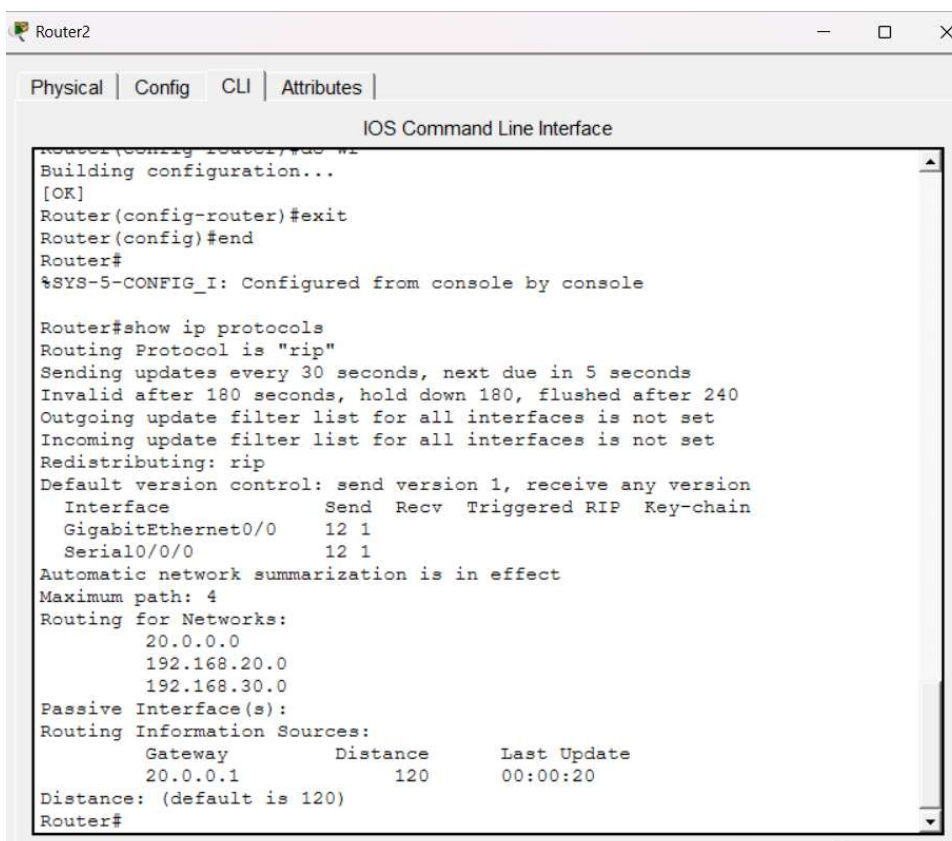


Figure 3: Router2 – show ip protocols

Router0, Router1 and Router2 show “Default version control: send version 2, receive 2”, confirming RIPv2 is correctly enabled and each interface's Send/Recv column reads “2 2”.

Task 3: Change the hop count metric for one of the networks by adding an extra router between two existing routers, then observe and record how the hop count value changes in the routing tables.

This task needs to be carried out directly in your Packet Tracer file so the hop counts you record are your own real results, not assumed ones. Follow these steps:

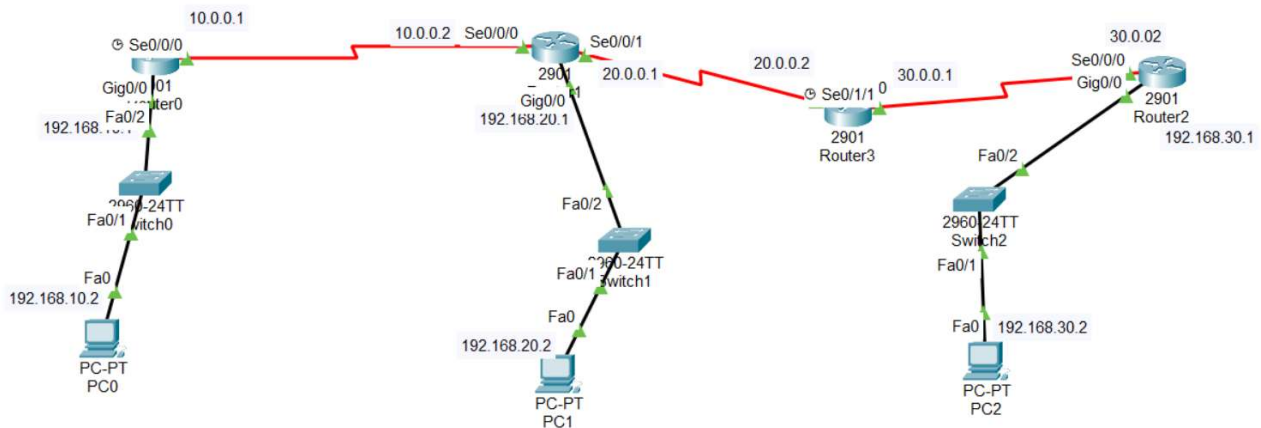
- Before making any change, we run show ip route on Router0 and note the current hop count for 192.168.30.0/24 (Router2's LAN) — it currently reads [120/2].

```
admin>enable
admin#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Serial0/0/0
L       10.0.0.1/32 is directly connected, Serial0/0/0
C       10.0.0.2/32 is directly connected, Serial0/0/0
R       20.0.0.0/8 [120/1] via 10.0.0.2, 00:00:06, Serial0/0/0
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
R       192.168.20.0/24 [120/1] via 10.0.0.2, 00:00:06, Serial0/0/0
R       192.168.30.0/24 [120/2] via 10.0.0.2, 00:00:06, Serial0/0/0
```

- A new router (e.g. Router3) added between Router1 and Router2 and routers IP address re-configured.



- Wait 30–60 seconds for RIP to reconverge, and “show ip route” command is re-run on Router0 and to check the hop count for 192.168.30.0/24 again — it has increased by 1, e.g. from [120/2] to [120/3]. Screen shot attached.

```

Router0
Physical | Config | CLI | Attributes |
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed
state to up

admin>enable
admin#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Serial0/0/0
L       10.0.0.1/32 is directly connected, Serial0/0/0
C       10.0.0.2/32 is directly connected, Serial0/0/0
R       20.0.0.0/8 [120/1] via 10.0.0.2, 00:00:15, Serial0/0/0
R       30.0.0.0/8 [120/2] via 10.0.0.2, 00:00:15, Serial0/0/0
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/1
R       192.168.20.0/24 [120/1] via 10.0.0.2, 00:00:15, Serial0/0/0
R       192.168.30.0/24 [120/3] via 10.0.0.2, 00:00:15, Serial0/0/0

admin#
Copy Paste
Top

```


Task 4: Use the simulation mode in Cisco Packet Tracer to capture and analyze RIP update packets as they are exchanged between routers. Hint: Filter for RIP packets in the simulation panel and note the source, destination, and advertised routes in each update.

PDU Information at Device: Router0

OSI Model | Inbound PDU Details

At Device: Router0
Source: Router1
Destination: 224.0.0.9

In Layers	Out Layers
Layer 7: RIP Version: 2, Command: 2	Layer7
Layer6	Layer6
Layer5	Layer5
Layer 4: UDP Src Port: 520, Dst Port: 520	Layer4
Layer 3: IP Header Src. IP: 10.0.0.2, Dest. IP: 224.0.0.9	Layer3
Layer 2: PPP Frame PPP	Layer2
Layer 1: Port Serial0/0/0	Layer1

1. The device receives a RIP RESPONSE.
2. The device examines all of the routes in the received RIP RESPONSE.
3. The device examines the route 20.0.0.0/8.
4. This route already exists in the RIP database. The device examines it to check if it needs to be updated.
5. The device examines the route 30.0.0.0/8.
6. This route already exists in the RIP database. The device examines it to check if it needs to be updated.
7. The device examines the route 192.168.20.0/24.
8. This route already exists in the RIP database. The device examines it to check if it needs to be updated.

Challenge Me << Previous Layer Next Layer >>

Multicast Destination: RIPv2 updates are delivered to the reserved IPv4 multicast address **224.0.0.9** rather than broadcast addresses (255.255.255.255), reducing CPU processing overhead on non-RIP devices

PDU Information at Device: Router0

OSI Model | Inbound PDU Details |

At Device: Router0
Source: Router1
Destination: 224.0.0.9

In Layers	Out Layers
Layer 7: RIP Version: 2, Command: 2	Layer7
Layer6	Layer6
Layer5	Layer5
Layer 4: UDP Src Port: 520, Dst Port: 520	Layer4
Layer 3: IP Header Src. IP: 10.0.0.2, Dest. IP: 224.0.0.9	Layer3
Layer 2: PPP Frame PPP	Layer2
Layer 1: Port Serial0/0/0	Layer1

1. The destination IP address is a broadcast or multicast address. The device dispatches the packet to the upper layer.

Challenge Me << Previous Layer Next Layer >>

Header / Field Level	Parameter	Value / Description
OSI Layer 7 (Application)	Protocol & Version	RIP Version 2 , Command: 2 (Response / Update)
OSI Layer 4 (Transport)	Ports	UDP Source Port: 520 , Destination Port: 520
OSI Layer 3 (Network)	Source & Destination IP	Source IP: 10.0.0.2 (Router1 Serial0/0/0 interface), Destination IP: 224.0.0.9 (RIPv2 Multicast)
OSI Layer 2 (Data Link)	Link Encapsulation	PPP Frame
OSI Layer 1 (Physical)	Interface	Serial0/0/0

When Router0 inspects the inbound RIP Response message from Router1, it processes the following advertised routes in its RIP database:

- **20.0.0.0/8** — Advertised network segment between Router1 and Router3.
- **30.0.0.0/8** — Advertised network segment between Router3 and Router2.
- **192.168.20.0/24** — Advertised LAN behind Router1.
- **192.168.30.0/24** — Advertised LAN behind Router2.